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PERFORMANCE EVALUATION OF CLOUD SERVERS

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Im this assignment the cloud service Provider is being considered.

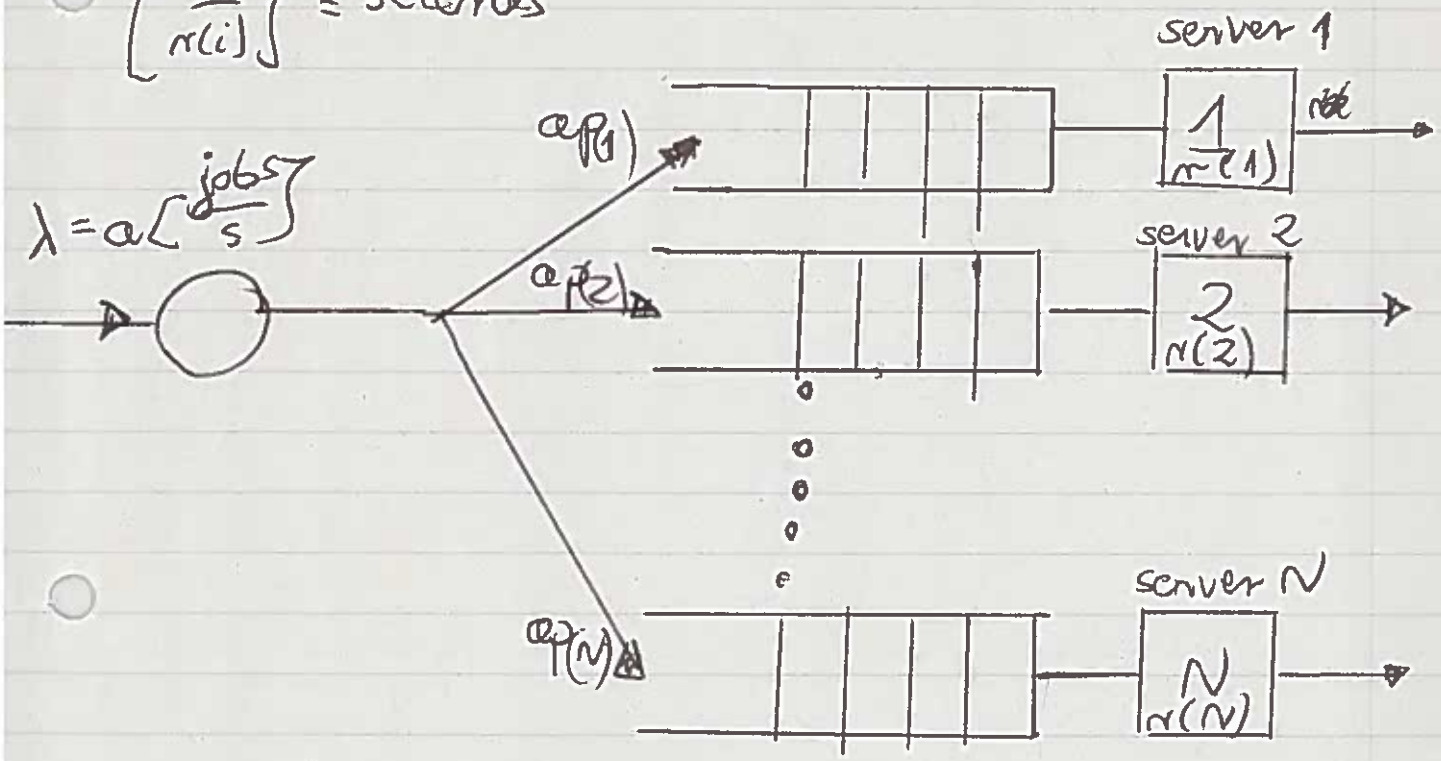
we assume that there are N servers

Each of the server ~~can be~~ has a FIFO service,

we assume that the service time is exponentially distributed.

The sevice time of rates. $r(1), \dots r(N)$

$\left[\frac{1}{r(i)}\right]$ = seconds



a: the rate at which the cloud service provicer receives jobs

$r(1), r(2), \dots, r(N)$: service time at each server

$p(1), p(2), \dots p(N)$: proportion at which jobs will be send to each server.

Herefore $ap(n)$ will represent the number of jobs send to the server 1

w: average waiting time.

The notation have to be introduced,

$k_i^+ = (k_1, k_2, \ldots, k_i + 1, k_{i+1}, \ldots, k_N)$

$k_i^- = (k_1, k_2, \ldots, k_i - 1, k_{i+1}, \ldots, k_N)$

$k = (k_1, k_2, \ldots, k_N)$

$1[k_i = 0]$: it equals to 1 when $k_i = 0$, and 0 otherwise

$1[k_i > 0]$: it is equal to 1 when $k_i > 0$, and 0 otherwise

Please ~~tell~~ note that above notations was used for the Assignment 3.

1) the joint probability distribution in steady state

$p(k) \sum_{i=1}^{N} (ap(i) + r(i)\, 1[k_i > 0])$ ①: this describes the situation when there are no arrival and no departures ⇒ queues will be the same length.

⇒ this actually represents all possible flows out of the steady state $P(k)$

$\sum_{i=1}^{N} p(k_i^+) r(i) + \sum_{i=1}^{N} p(k_i^-) ap(i)\, 1[k_i > 0]$ ②

this represents the situation when there is a departure from a server, and

equating ① and ②

$$p(k) \sum_{i=1}^{N} (\alpha p(i) + r(i) 1[k_i > 0]) = \sum_{i=1}^{N} p(k_i^+) r(i) +$$

$$+ \sum_{i=1}^{N} p(k_i^-) \alpha p(i) 1[k_i > 0]$$

$$\Updownarrow$$

$$P(n(1), n(2), \ldots, n(N)) \overset{②}{\sum_{i=1}^{N}} (\alpha p(i) + r(i) 1[n(i) > 0]) =$$

$$= \sum_{i=1}^{N} \overset{②}{P(n(i)^+)} r(i) + \sum_{i=1}^{N} \overset{③}{P(n(i)^-)} \alpha p(i) 1[n(i) > 0]$$

(a.) there is a situation where all the queues are the same i.e same length.

It can be said that the total number of packets in each of the queues remains the same $\Rightarrow$ this can happen when there are no arrivals and no departures

(b) there is a departure from the i-th server in i-th server there are $n[i]+1$ jobs penalty

(c) an arrival happened to the server i

there are $n(i)-1$ jobs at i-th server pending

$$p(n(1), n(2), \dots, n(N)) \sum_{i=1}^{N} (a\,p(i) + r(i)\,1[n(i)>0]) =$$

$$= \sum_{i=1}^{N} p(n(i)^+) r(i) + \sum_{i=1}^{N} p(n(i)^-) a\,p(i)\,1[n(i)>0]$$

$$\Downarrow$$

$$p(n(1), n(2), \dots, n(N)) = \frac{\sum_{i=1}^{N} p(n(i)^+) r(i) + \sum_{i=1}^{N} p(n(i)^-) a\,p(i)\,1[n(i)>0]}{\sum_{i=1}^{N} (a\,p(i) + r(i)\,1[n(i)>0])}$$

the above expression represents the joint probability distribution.

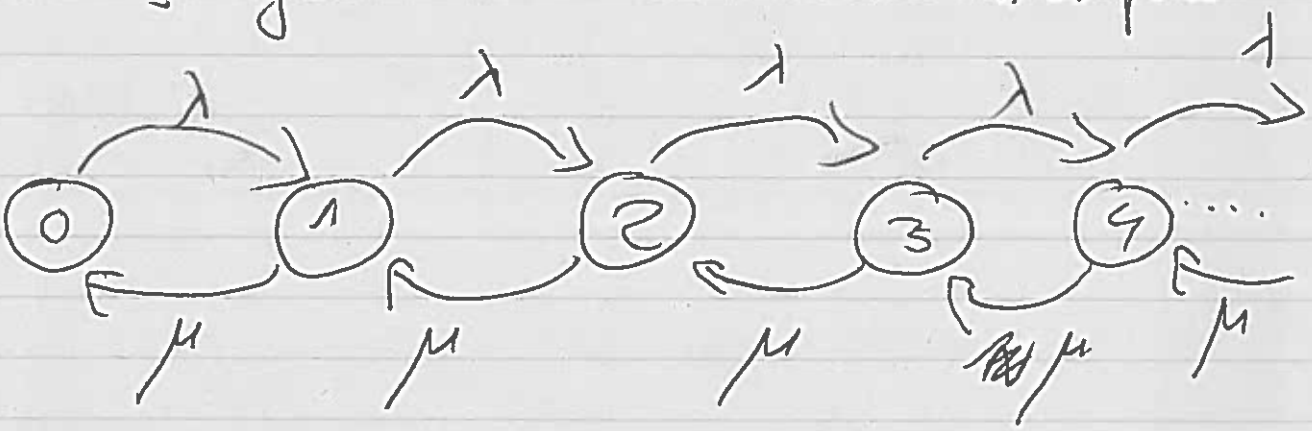
2) The average response time W of jobs arriving to the system

As it was mentioned on the -1- page the system can be modelled as a combination of independent M/M/1 systems

Moreover its assumed that arrival and departure rates are different.

The balanced equations can be derived from the state-transition diagram.

For 1 system the state-transition diagram:



At (0) where P_0 represents the probability of being in state 0

$$\mu P_1 = \lambda P_0$$

$$\Downarrow$$

$$P_1 = \frac{\lambda P_0}{\mu}$$

At (1)

$$P_1(\mu+\lambda) = \lambda P_0 + \mu P_2 \qquad \text{and} \quad P_1 = \frac{\lambda P_0}{\mu}$$

$$\Downarrow$$

$$\mu_2 P_2 = \frac{\lambda}{\mu} P_0(\mu+\lambda) - \lambda P_0$$

$$\mu P_2 = \frac{\lambda}{\mu} P_0 \mu + \frac{\lambda^2}{\mu} P_0 - \lambda P_0$$

$$\mu P_2 = \cancel{\lambda P_0} + \frac{\lambda^2}{\mu} P_0 - \cancel{\lambda P_0}$$

$$\mu P_2 = \frac{\lambda^2}{\mu} P_0 \;/\!:\mu$$

$$\boxed{P_2 = \frac{\lambda^2}{\mu^2} P_0}$$

At (2)

$$P_2(\mu+\lambda) = \lambda P_1 + \mu P_3$$

$$\Downarrow$$

$$\frac{\lambda^2}{\mu^2} P_0(\mu+\lambda) = \frac{\lambda^2 P_0}{\mu} + \mu P_3$$

and $P_2 = \frac{\lambda^2}{\mu^2} P_0$ and $P_1 = \frac{\lambda P_0}{\mu}$

$$\mu P_3 = \frac{\lambda^2}{\mu^2} P_0(\mu+\lambda) - \frac{\lambda^2}{\mu} P_0$$

$$\mu P_3 = \frac{\lambda^2}{\mu^2} P_0 \mu + \frac{\lambda^3}{\mu^2} P_0 - \frac{\lambda^2}{\mu} P_0$$

$$\mu P_3 = \frac{\lambda^2}{\mu} P_0 - \frac{\lambda^2}{\mu} P_0 + \frac{\lambda^3}{\mu^2} P_0$$

$$\Downarrow$$

$$\boxed{P_3 = \frac{\lambda^3}{\mu^3} P_0}$$

it can be seen that there is a pattern

therefore the probability of being in state k can be related with probability of being in state 0.

$$p_k = \left(\frac{\lambda}{\mu}\right)^k p_0$$

Please note that being in state k can be seen as having k jobs in the system

For a notation purposes $\frac{\lambda}{\mu} = \rho$

therefore

$$\boxed{p_k = \rho^k p_0}$$

by using above statement its possible to calculate the average number of jobs in the System

$$E\{N_i\} = \sum_{k=1}^{\infty} k p_k = \sum_{k=1}^{\infty} k \rho_i^k (1-\rho_i) = \frac{(1-\rho)\rho}{(1-\rho)^2} =$$

$$= \frac{\rho}{1-\rho} = (1-\rho_i) \sum_{k=1}^{\infty} k \rho_i^k = \frac{(1-\rho_i)\rho_i}{(1-\rho_i)^2} = \boxed{\frac{\rho_i}{1-\rho_i}}$$

the average number of jobs $= \frac{\rho_i}{1-\rho_i}$

The average response time can be obtained by using the Little's Theorem.

$$E[W_i] = \frac{E[N_i]}{\lambda_i} = \frac{\frac{\lambda_i}{\mu_i}}{(1-\frac{\lambda_i}{\mu_i})(\lambda_i)} = \frac{\frac{\lambda_i}{\mu_i}}{\frac{\mu_i-\lambda_i}{\mu_i}\lambda_i} =$$

$$= \frac{\lambda_i}{\mu_i} \cdot \frac{\mu_i}{(\mu_i-\lambda_i)\lambda_i} = \frac{1}{\mu_i-\lambda_i}$$

By substituting for $\mu_i \Rightarrow r(i)$ and $\lambda_i \Rightarrow a_{p(i)}$ (as defined on the first page) we obtain

$$E[W_i] = \frac{1}{r(i) - a_{p(i)}}$$

above statement describes the average response time in i-th server. therefore to find the response time of the entire system we can calculate the expected value.

$$E\{w\} = \sum_{i=1}^{N} p(i) w_i = \sum_{i=1}^{N} \frac{p(i)}{r(i) - a p(i)}$$

3) in order to minimize w the $p(i)$ should be optimized

$$E\{w\} = \sum_{i=1}^{N} p(i) w_i = \sum_{i=1}^{N} \frac{p(i)}{r(i) - a p(i)} \quad /\!/ \cdot r(i)$$

$$\Rightarrow E\{w\} = \sum_{i=1}^{N} \frac{p(i)}{r(i) - a p(i)} = \sum_{i=1}^{N} \frac{\frac{p(i)}{r(i)}}{\frac{r(i)}{r(i)} - \frac{a p(i)}{r(i)}} =$$

$$= \sum_{i=1}^{N} \frac{\frac{p(i)}{r(i)}}{1 - \frac{a p(i)}{r(i)}} \quad *$$

For a notation reasons let's equate $\frac{1}{r(i)}$ to $E\{S_i\}$ (mean service at i-th service)

By applying the above notation * becomes

$$\sum_{i=1}^{N} \frac{p(i) E\{S_i\}}{1 - a p(i) E\{S_i\}}$$

$$\sum_{i=1}^{N} \frac{p(i)E\{S_i\}}{1-\alpha p(i)E\{S_i\}}$$

By expanding above terms we obtain

$$\sum_{i=1}^{N} \frac{p(i)E\{S_i\}}{1-\alpha p(i)E\{S_i\}} = \frac{p(1)E\{S_1\}}{1-\alpha p(1)E\{S_1\}} + \sum_{i=2}^{N} \frac{p(i)E\{S_i\}}{1-\alpha p(i)E\{S_i\}} \quad **$$

$$p(1)+p(2)+\ldots+p(N) = 1$$

$$p(1)+\left[p(2)+\ldots+p(N)\right] = 1$$

$$p(1) = 1-\left[p(2)+\ldots+p(N)\right]$$

$$\boxed{p(1) = 1-\sum_{i=2}^{N} p(i)}$$

substituting above to **

$$E\{W\} = \frac{\left(1-\sum_{i=2}^{N} p(i)\right)E\{S_1\}}{1-\alpha\left(1-\sum_{i=2}^{N} p(i)\right)E\{S_1\}} + \sum_{i=2}^{N} \frac{p(i)E\{S_i\}}{1-\alpha p(i)E\{S_i\}} \quad ***$$

in order to calculate to $p(i)$ that minimizes W the derivative of *** has to be taken

$\frac{\partial E[w]}{\partial p(i)} \Rightarrow (E[w])'$ simplified notation

$$(E[w])' = \left(\frac{\overbrace{\left[1 - \sum_{i=2}^{N} p(i)\right] E[S_1]}^{①}}{1 - a\left(1 + \sum_{i=2}^{N} p(i)\right) E[S_1]} \right)' + \left(\sum_{i=2}^{N} \overbrace{\frac{p(i) E[S_i]}{1 - a p(i) E[S_i]}}^{②} \right)'$$

$$① \left(\frac{\left[1 - \sum_{i=2}^{N} p(i)\right] E[S_1]}{1 - a\left(\sum_{i=2}^{N} p(i)\right) E[S_1]} \right)' = \overbrace{\left(1 - \sum_{i=2}^{N} p(i) E[S_1]\right)'}^{(*)} \times$$

$$\times \left(\frac{1 - a\left(1 - \sum_{i=2}^{N} p(i)\right) E[S_1]}{\left(1 - \sum_{i=2}^{N} p(i) E[S_1]\right)\left(1 - a\left(1 - \sum_{i=2}^{N} p(i)\right) E[S_1]\right)'} \right)' \overbrace{\quad}^{(**)} =$$

$$= \left(\frac{-E[S_1] \cdot \left(1 - \left(a + a \sum_{i=2}^{N} p(i)\right) E[S_1]\right)}{\left(1 - \sum_{i=2}^{N} p(i) E[S_1]\right)\left(1 - a\left(1 - \sum_{i=2}^{N} p(i)\right) E[S_1]\right)'} \right) =$$

$$= \left(\frac{(-E[S_1])(1 - a E[S_1] + a E[S_1] \sum_{i=2}^{N} p(i))}{\left(1 - \sum_{i=2}^{N} p(i)\right) E[S_1]\left(1 - a\left(1 - \sum_{i=2}^{N} p(i)\right) E[S_1]\right)'} \right) (*)$$

$$(**) \left(\left(1 - \sum_{i=2}^{N} p(i)\right) E[S_1]\right)\left(1 - a\left(1 - \sum_{i=2}^{N} p(i)\right) E[S_1]\right)'$$

$$\left(1 - \sum_{i=2}^{N} p(i)\right)(E[S_1])(a E[S_1])$$

$$(*) + (**) : \left(-E[S_1](1 - a E[S_1]) + a E[S_1] \sum_{i=2}^{N} p(i)\right) -$$

$$- \left(1 - \sum_{i=2}^{N} p(i)\right)(E[S_1])(a E[S_1]) \quad i = 2$$

please note that the denominator t^2

$= \left(1 - a\left(1 - \sum_{i=2}^{N} p(i) E[S_{1,J}]\right)\right)^2$ for simplification

notation it's called t^2

$$\textcircled{2}\left(\sum_{i=2}^{N} \frac{p(i) E[S_{i,J}]}{1 - a p(i) E[S_{i,J}]}\right)' =$$

$$\left((p(i)E[S_{i,J}]\right)'(1 - a\,p(i)E[S_{i,J}]) - (p(i)E[S_{i,J}])\cdot$$

$$\cdot(1 - a p(i) E[S_{i,J}]) : u^2 \Rightarrow$$

$$u^2 = (1 - a\,p(i)E[S_{i,J}])^2$$

$$\Rightarrow [E[S_{i,J}](1 - a p(i) E[S_{i,J}]) - (p(i) E[S_{i,J}]) *$$

$$* (-a E[S_{i,J}])] // : u^2$$

Combining ① and ②

$$[(-E[S_{1,J}])(1 - aE[S_{1,J}] + aE[S_{1,J}] \sum_{i=2}^{N} p(i))) -$$

$$-(1 - \sum_{i=2}^{N} p(i)](E[S_{1,J}])(aE[S_{1,J}])] \cdot t^{-2} +$$

$$+[(E[S_{i,J}])(1 + a p(i) E[S_{i,J}]) - (p(i) E[S_{1,J}]) *$$

$$* (-a E[S_{i,J}])] u^{-2}$$

please note that this statement holds for $i \in \langle 2, N \rangle$

For the simplicity the denominators will be omitted for now, the terms will be modified and at the end divided by corresponding denominator

$$\left(-E[S_1]\right)\left(1 - aE[S_1] + aE[S_1]\sum_{i=2}^{N} p(i)\right) - \left(1 - \sum_{i=2}^{N} p(i)\right)\left(E[S_1]\left(aE[S_1]\right)\right)\Big)_0^{\infty} t^2$$

Lets call the ~~numerator~~ numerator m

$$m = \left[-E[S_1]\right]\left[1 - E[S_1]a\left(1 - \sum_{i=2}^{N} p(i)\right) - \left(1 - \sum_{i=2}^{N} p(i)\right)a\left(E[S_1]\right)^2\right.$$

similarly lets call the numerator of second term $\underline{\underline{m}}$

$\underline{\underline{\text{i.e.}}}$

$$\underline{\underline{m}} = \left[E[S_i]\right)\left(1 - ap(i)E[S_i]\right) - \left(p(i)E[S_i]\right) \times \left(-aE[S_i]\right)$$

$$m = E[S_i] - \left(E[S_i]\right)^2 ap(i) + p(i)\left(E[S_i]\right)^2 a$$

please note that

$$m = E[S_i^2] - (E[S_i])^2 \alpha p(i) + p(i)(E[S_i])^2 \alpha$$

$$m = E[S_i^2]$$

Recall that

$$\alpha \quad p(1) = 1 - \sum_{i=2}^{N} p(i)$$

substituting above into m

$$n = -[E[S_1^2][1 - E[S_1]\alpha p(1)] - p(1)\alpha(E[S_1])^2$$

$$n = -E[S_1^2] + (E[S_1])^2 \alpha p(1) - p(1)\alpha(E[S_1])^2$$

$$n = -E[S_1^2]$$

Recall

$$d^2 = (1 - \alpha p(i) E[S_i])^2$$

$$t^2 = (1 - \alpha(1 - \sum_{i=2}^{N} p(i)) E[S_1])^2$$

$$\Downarrow$$

$$t^2 = (1 - \alpha(p(1)E[S_1])^2$$

$$\frac{n}{t^2} + \frac{m}{d^2} = (E[W])'$$

therefore

$$\frac{-E[S_1^2]}{(1-\alpha(p(1)E[S_1])^2} + \frac{E[S_i^2]}{(1-\alpha p(i)E[S_i])^2} = (E[W])'$$

By assumption

$$\frac{1}{N(i)} = E[S_i] \quad \text{and} \quad \rho_i = \frac{a p(i)}{N(i)} \quad *$$

$$\Rightarrow \rho_i = \frac{a p(i)}{N(i)} = a p(i) E[S_i] \quad **$$

$$\frac{-E[S_1]}{(1 - a p(1) E[S_1])^2} \quad \frac{1}{(1 - a p(i) E[S_i])^2} \quad \frac{E[S_i]}{} = (E[W])'$$

substituting $*$ $*$ into above expressions

$$\frac{E[S_i]}{(1-\rho_i)^2} + \frac{-E[S_1]}{(1 - a p(1) E[S_1])^2}$$

$$\Downarrow (1)$$

$$(1-\rho_1)^2 \quad \text{(substituting } i=1 \text{ to } **)$$

$$\Downarrow$$

$$\frac{E[S_i]}{(1-\rho_i)^2} + \frac{-E[S_1]}{(1-\rho_1)^2} = (E[W])'$$

please note the condition $i \in \{2, \dots, N\}$ still holds

$(E[W])' = 0 \Rightarrow$

$$\frac{E[S_i]}{(1-\rho_i)^2} = \frac{E[S_1]}{(1-\rho_1)^2}$$

$$E[S_i](1-\rho_1)^2 = E[S_1](1-\rho_i)^2$$

$$\Rightarrow (1-\rho_1)^2 = \frac{E[S_1](1-\rho_i)^2}{E[S_i]}$$

(11)

$$\frac{(1-\rho_1)^2}{E[S_1]} = \frac{(1-\rho_i)^2}{E[S_i]} \quad *$$

By using $\frac{1}{n(i)} = E[S_i]$

* becomes

$$(1-\rho_1)^2 n(1) = (1-\rho_i)^2 n(i)$$

$$\frac{(1-\rho_1)^2 n(1)}{n(i)} = (1-\rho_i)^2$$

$$(1-\rho_i)^2 = \frac{(1-\rho_1)^2 n(1)}{n(i)} \quad /\sqrt{\ }$$

all the terms are ≥ 0 therefore dont need to take an absolute value

$$1-\rho_i = (1-\rho_1)\sqrt{\frac{n(1)}{n(i)}}$$

$$-\rho_i = -1 + (1-\rho_1)\sqrt{\frac{n(1)}{n(i)}} \quad // \times(-1)$$

$$\rho_i = 1 - (1-\rho_1)\sqrt{\frac{n(1)}{n(i)}}$$

$$\rho_i = 1 - (1-\rho_1)\sqrt{\frac{n(1)}{n(i)}}$$

By using

$$\rho_i = \frac{a\,p(i)}{n(i)}$$

$$\frac{a\,p(i)}{n(i)} = 1 - (1-\rho_1)\sqrt{\frac{n(1)}{n(i)}} \quad /\!/ \times n(i)$$

$$a\,p(i) = n(i) - (1-\rho_1)\sqrt{n(1)\,n(i)} \quad /\!/ \div a$$

$$p(i) = \frac{n(i) - (1-\rho_1)\sqrt{n(1)\,n(i)}}{a} \qquad i \in (2, N)$$

above $p(i)$ minimizes ω.

4 the average number of jobs $\bar{y}(i)$ at server i

please recall that

$$w_i = \frac{1}{r(i) - a p(i)}$$

By applying the Littles theorem

$$\bar{y}(i) = \lambda_i w_i$$

also $\lambda_i = a\,P(i) \Rightarrow$

$$\Rightarrow \bar{y}(i) = a P(i) w_i \Rightarrow \boxed{\bar{y}(i) = \frac{a P(i)}{r(i) - a p(i)}}$$

$p(i)$ obtained in the previous question 3

$$p(i) = \frac{r(i) - (1 - p_1)\sqrt{r(1) r(i)}}{a}$$

$y(i)$ becomes

$$y(i) = \frac{\alpha \cdot r(i) - (1-\rho_1)\sqrt{r(1)r(i)}}{\alpha}$$

$$\frac{}{r(i) - \alpha\left(r(i) - (1-\rho_1)\sqrt{r(1)r(i)}\right)}$$

$$y(i) = \frac{r(i) - (1-\rho_1)\sqrt{r(1)r(i)}}{r(i) - \left(r(i) - (1-\rho_1)\sqrt{r(1)r(i)}\right)} \Rightarrow$$

$$\Rightarrow \frac{r(i) - (1-\rho_1)\sqrt{r(1)r(i)}}{r(i) - r(i) + (1-\rho_1)\sqrt{r(1)r(i)}} =$$

$$= \frac{r(i)}{(1-\rho_1)\sqrt{r(1)r(i)}} - 1$$

5. A practical heuristic rule for dispatching jobs to the servers so that the optimum W^* of W may be obtained

It is known that in order to have a stable system $|\lambda_i < \mu_i|$

From the above

$$a_p(i) < r(i)$$

It is assumed that each of the servers is different and therefore it can be written as:

$$r(1) \neq r(2) \neq \ldots \neq r(N)$$

Lets consider the situation in which the service times of nodes are decreasing with the server index i.e. The server 1 has the highest value of service rate

we can sort $r(i)$ from the fastest to the slowest obtaining

$$r(1) > r(2) > \ldots > r(N)$$

it can be observed that

$$a_p(i) < r(i)$$

The number of packets sent to the i-th

serves ~~should~~ have to be smaller than r(i) *

Let us consider a simple example

r(1) is high

p(1) = 0.1

r(2) < r(1)

p(2) = 0.9

server 2 is being overloaded while the server 1 still could accept packets. thus conflicts with (*)

therefore it can be said by using * and the "sorted" service rates that,

$$\begin{cases} \rho(i) < r(i) \\ r(1) > r2 > \dots > r(N) \end{cases}$$

It can be said that the server with higher service rate should obtain more jobs per second, than the server with small r(i)

server have to be smaller than $f(i)$

Let us consider a simple example

$r(1)$ is high

$p(1) = 0.1$

$N(2) < r(1)$

$p(2) = 0.9$

server 2 is being overloaded while server 1 still could accept packets

Therefore it can be said

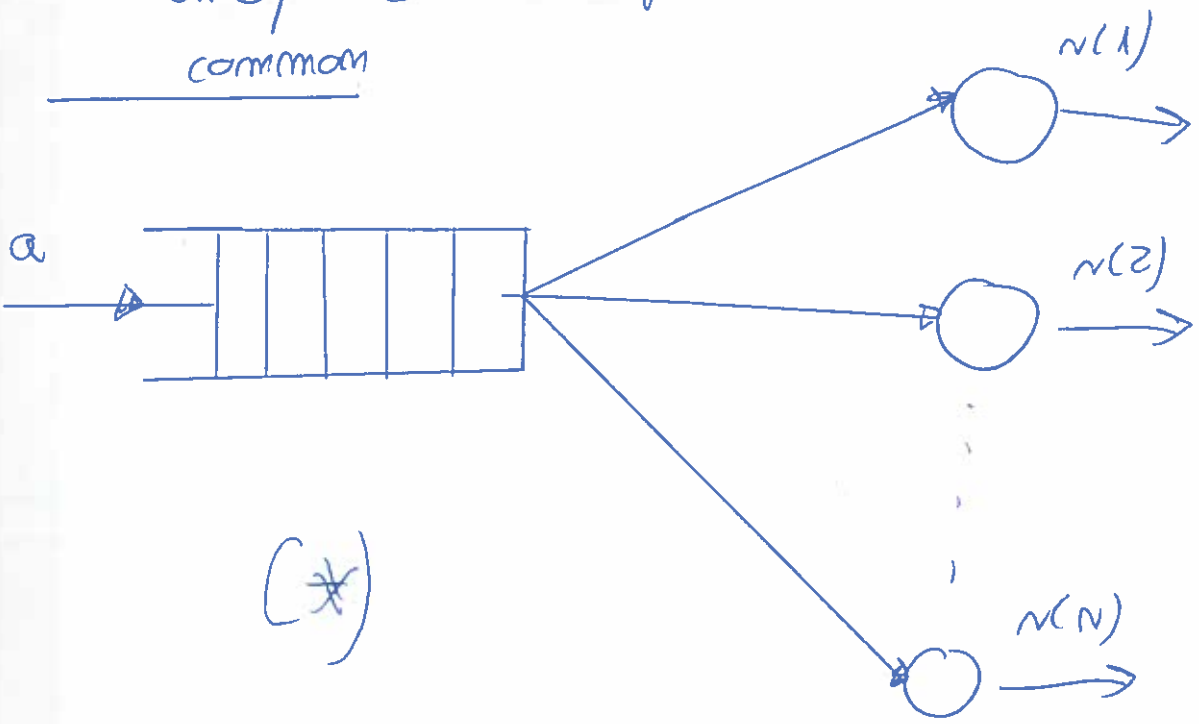
$a(p(i)) < r(i)$

while

$r(1) > r(2) > \dots > r(N)$

Server with higher service rate should obtain more packets by
It can ~~perfor~~ deal with more jobs per second than the server with small $r(i)$

Question 6 | Comparison of systems with one and multiple queues common



In the above system it can be seen that there is a single queue. Therefore the incoming packets are "waiting" in this single queue before being send to servers.

It differs from the first system because packets which are being sent to the servers are being processed immediately because there are no separate queues for each of the system.

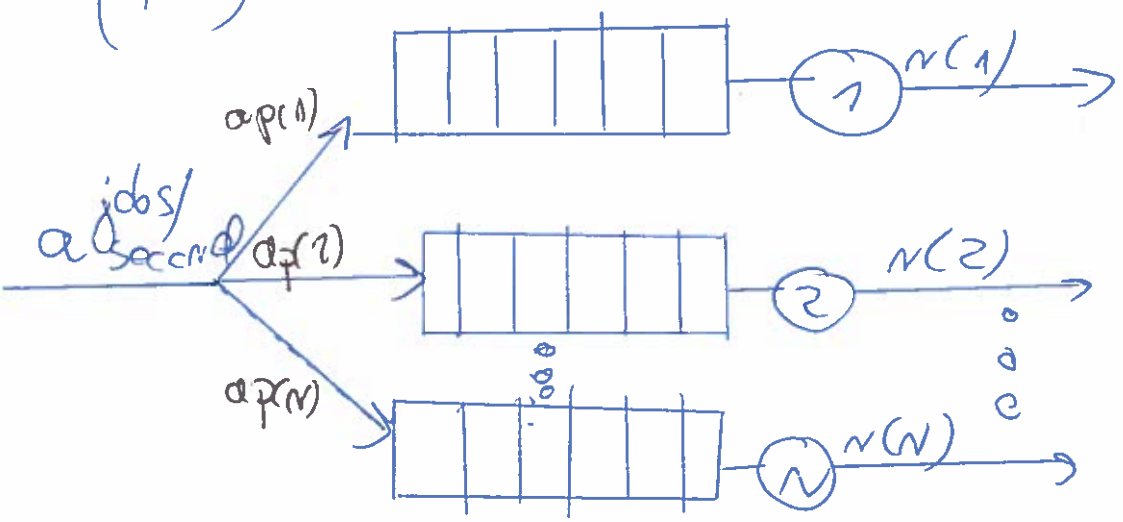
Incoming packets are being queued if there is at least 1 free server packets will go to the "free" server.

From the other hand if there is more than one server which is not busy (idle) then the packets will be send to this server which has the fastest service rate $\left(\frac{1}{r(i)}\right)$

Please a note that the total arrival rate for the system above is the same as the system compared in question 1

The multiple queue system from the question 1

(**)



when the two systems are compared there have to be two separate scenarios.

(1) server has broken

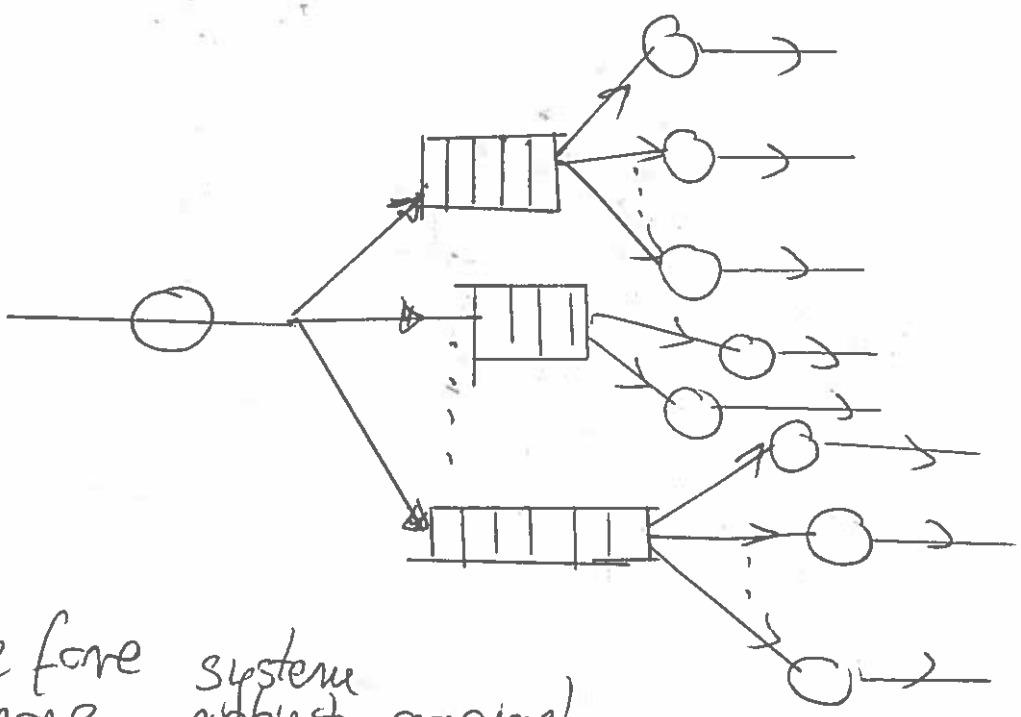
(2) queue has broken

in the (1) situation that the 1 server has broken the system (*) only packets which were processed are lost Traffic from the queue will not be forwarded to the "broken server branch". the total number of packets processed in the system in the unit time is obviously lower because there are N−1 servers available.

For the System (**) if one of the servers is broken there are all the packets lost which queued in this servers queue. Because there is no connection between queues packets from the "broken branch can not be forwarded to the other branch.

Another thing about (**) is that there is a chance of one server being idle (because (r(i)) is very high while there are other packets being queued as mentioned before there is no connection between queues and hence the traffic cannot be rerouted. And it is this situation one of the server is not doing anything while others are busy processing incoming packets

The ideal situation would be a combination of (*) and (**)



therefore system is more robust against the queue failure and the server breakdown. Please note that above system has its own drawbacks, however in terms of reliability is better than (*) and (**)

One the other hand if (2) happens that is the queue is broken the system(*) there are all the packets being lost. Please note that this means the system is not being able to operate any longer. the queue is a critical point of (*)

For the (**) only the packets which were queued in 1 queue are lost. System can operate further.

In order to make (*) "workable" again the whole system must be restarted while for the queue failure (**) operates with lower efficiency.

Another difference while these 2 approaches are compared is a price. Obviously the multiqueued system is more expensive, however it is more robust⇒ the server and queue failures have similar impact of losing packets being in the queue.

For those System in (**) there may be situation that service rate is small but there is a tiny probability of packets coming to that branch. Therefore queue length will be constantly increasing. to server while in (*) packets would be forwarded and queue length would decrease

overall it can be concluded that system (*) has more advantages than disadvantages. Although it is sensitive to a queue break down it is robust for a server break down.
Another advantage is that the packets are being constantly send to the first idle server therefore queue is constantly decreasing. Another advantage is a system cost. which is lower than system (**)

By analyzing (*) and (**) it can be said that both systems can be improved by applying the acknowledgement.
If for some reason there is not acknowledgment after a certain time. the Job dispatcher in (**) can change the p-value so that there are no packets being sent to the "broken" branch ie. The branch from there was not an acknowledgement.

[1] L. Wang, E. Gelenbe, "Adaptive Dispatching of Tasks in the Cloud", IEEE Transactions on Cloud Computing, 2015